

# **Customer Churn Prediction and Retention Intelligence System**

End-to-End Machine Learning Project  
Internship Evaluation Submission

## 1. Abstract

This project presents a complete end-to-end machine learning system for predicting customer churn in a telecommunications company. Using the publicly available Telco Customer Churn dataset from Kaggle (7,043 customers, 21 features), the system implements data cleaning, exploratory data analysis, business KPI calculation, feature engineering, multiple classification models, hyperparameter tuning, customer risk segmentation, and a rule-based business recommendation engine. The final XGBoost (Tuned) model achieves ~84% ROC-AUC and ~81% Recall.

## 2. Business Problem

The business wants to identify customers who are at risk of leaving and answer:

- What is the overall churn rate?
- Which customer groups have higher churn?
- What factors are associated with churn?
- Which customers are predicted to be at high risk?

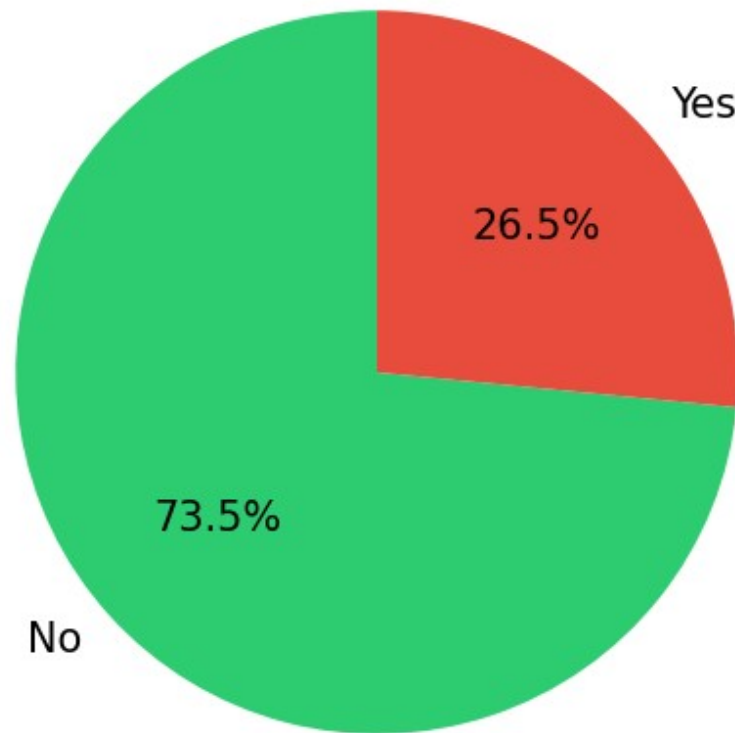
## 3. Dataset Description

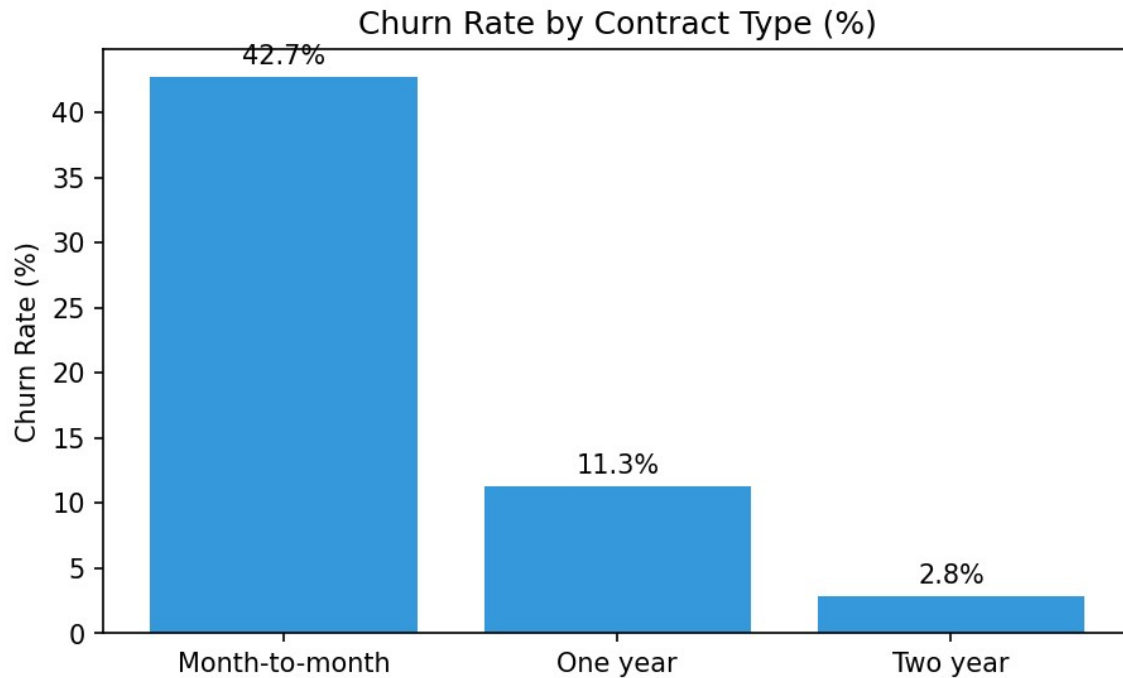
| Property            | Value                                    |
|---------------------|------------------------------------------|
| Rows                | 7,043                                    |
| Columns             | 21                                       |
| Target Column       | Churn (Yes/No)                           |
| Target Distribution | No: 5,174 (73.46%)   Yes: 1,869 (26.54%) |

## 4. Exploratory Data Analysis

Comprehensive EDA was performed to understand churn patterns:

## Customer Churn Distribution



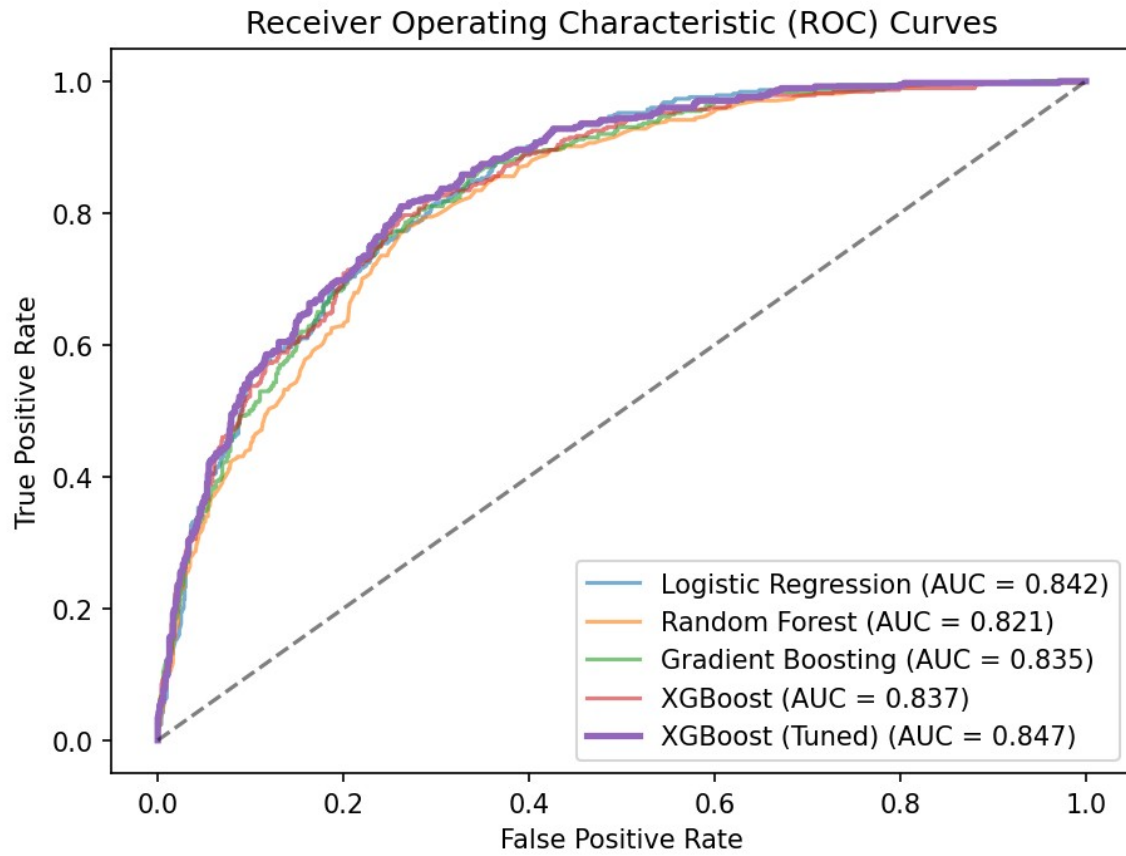


| Analysis         | Finding                                                        |
|------------------|----------------------------------------------------------------|
| Overall Churn    | 26.54% churn rate (1,869 out of 7,043)                         |
| Contract Type    | Month-to-month: 42.7% churn   One year: 11.3%   Two year: 2.8% |
| Internet Service | Fiber optic: 41.9%   DSL: 19.0%   No internet: 7.4%            |
| Payment Method   | Electronic check: 45.3% (highest)                              |

## 5. Machine Learning Models

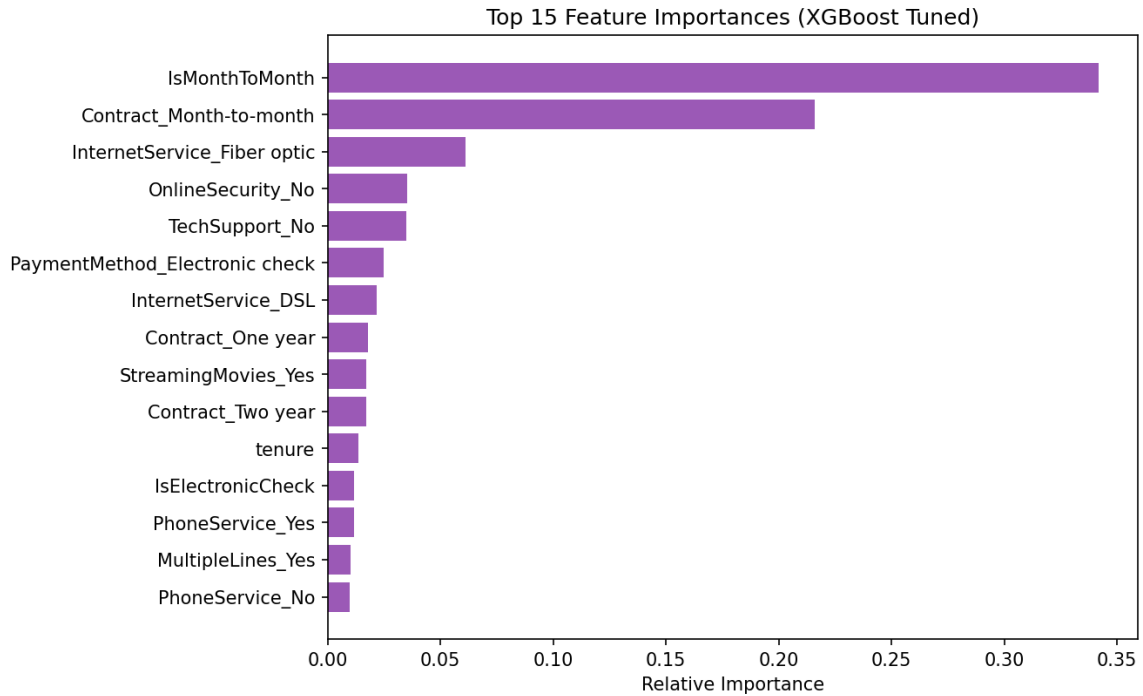
| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC |
|---------------------|----------|-----------|--------|--------|---------|
| Logistic Regression | 74.24%   | 50.96%    | 78.34% | 61.75% | 84.20%  |
| Random Forest       | 77.71%   | 60.27%    | 47.06% | 52.85% | 82.12%  |
| Gradient Boosting   | 79.21%   | 63.11%    | 52.14% | 57.10% | 83.52%  |
| XGBoost             | 75.51%   | 52.64%    | 77.27% | 62.62% | 83.72%  |
| XGBoost (Tuned)     | 75.66%   | 52.70%    | 81.02% | 63.86% | 84.71%  |

ROC Curves of all trained models:



## 6. Feature Importance

The final XGBoost model identified the following features as most predictive of churn:



## 7. Business Insights & Recommendations

| Condition                    | Action                                                             |
|------------------------------|--------------------------------------------------------------------|
| High risk + MTM contract     | Offer incentivised annual or 2-year contract upgrade               |
| High risk + Short tenure     | Proactive onboarding check-in, welcome discount, dedicated support |
| High risk + High charges     | Personalised bundle offers or loyalty discounts                    |
| High risk + Fiber optic      | Investigate service quality; offer proactive technical support     |
| High risk + Electronic check | Encourage auto-pay with a small incentive                          |